and $H_2O(l)$. The enthalpies of formation are –277, – 393.5 and –285.5 kJ mol^{-1} respectively.

$$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$$

$$\Delta H_r^0 = \left\{\left(\Delta H_f^0\right)_{products} - \left(\Delta H_f^0\right)_{reactants}\right\}$$

$$\Delta H_r^0 = \left[2\left(\Delta H_f^0\right)_{CO_2} + 3\left(\Delta H_f^0\right)_{H_2O}\right] - \left[1\left(\Delta H_f^0\right)_{C_2H_5OH} + 3\left(\Delta H_f^0\right)_{O_2}\right]$$

$$\Delta H_r^0 = \begin{bmatrix} 2mol\,(-393.5)\,kJ\,mol^{-1} \\ +\,3mol\,(-285.5)\,kJ\,mol^{-1} \end{bmatrix} - \begin{bmatrix} 1mol\,(-277)\,kJ\,mol^{-1} \\ +\,3mol\,(0)kJ\,mol^{-1} \end{bmatrix}$$

$$= [-787 - 856.5] - [-277]$$

$$= -1643.5 + 277$$

$$\Delta H_r^0 = -1366.5\ KJ$$

Evaluate Yourself - 1

Calculate ΔH_f^0 for the reaction

$CO_2(g) + H_2(g) \rightarrow CO(g) + H_2O(g)$

given that ΔH_f^0 for CO_2 (g), CO (g) and H_2O(g) are – 393.5, – 111.31 and – 242 kJ mol^{-1} respectively.

Heat of combustion

The heat of combustion of a substance is defined as "The change in enthalpy of a system when one mole of the substance is completely burnt in excess of air or oxygen". It is denoted by ΔH_C. For example, the heat of combustion of methane is – 87.78 kJ mol^{-1}

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

ΔH_C = – 87.78 kJ mol^{-1}

For the combustion of carbon,

$C(s) + O_2(g) \rightarrow CO_2(g)$

ΔH_C = – 394.55 kJ mol^{-1}

Combustion reactions are always exothermic. Hence the enthalpy change is always negative.

Molar heat capacities

When heat (q) is supplied to a system, the molecules in the system absorb the heat and hence their kinetic energy increases, which in turn raises the temperature of the system from T_1 to T_2. This increase $(T_2 - T_1)$ in temperature is directly proportional to the amount of heat absorbed and inversely proportional to mass of the substance. In other words,

$q \alpha m\Delta T$

$q = c\ m\ \Delta T$

$c = q/m\ \Delta T$

The constant c is called heat capacity.

$$c = \left(\frac{q}{m\,(T_2 - T_1)}\right) \quad -------(7.18)$$

when m=1 kg and $(T_2 - T_1)$ = 1 K then the heat capacity is referred as specific heat capacity. The equation 7.18 becomes

$c = q$

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